

Total marks (120)**Attempt questions 1 – 10****All questions are of equal value**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)	Use a SEPARATE writing booklet	Marks
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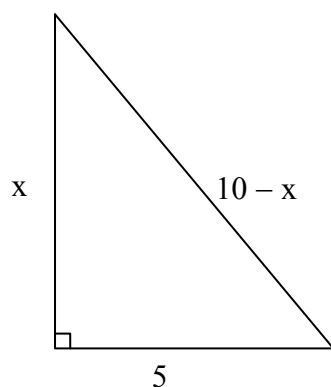
- | | | |
|-----|---|---|
| (a) | Find the value of $4\pi\sqrt{\frac{a}{g}}$ where $a = 4.2$ and $g = 9.8$. Give your answer correct to 2 significant figures. | 2 |
|-----|---|---|

- | | | |
|-----|---|---|
| (b) | Solve the pair of simultaneous equations
$x + y = 2$
$2x - y = 7$ | 2 |
|-----|---|---|

- | | | |
|-----|----------------------------|---|
| (c) | Factorise $2x^2 - 3x - 14$ | 1 |
|-----|----------------------------|---|

- | | | |
|-----|--|---|
| (d) | Write down the exact value of 135° in radians | 1 |
|-----|--|---|

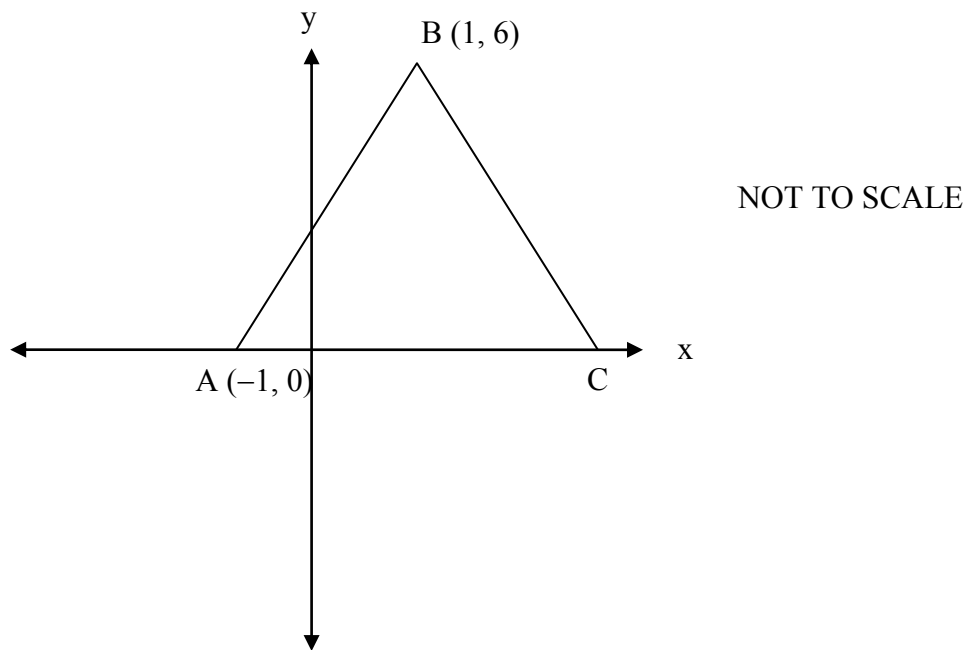
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|-----|---|---|
| (e) | Find the value of x in the right angled triangle. | 2 |
|-----|---|---|



- | | | |
|-----|--|---|
| (f) | Rationalise the denominator $\frac{2}{\sqrt{5} - \sqrt{3}}$. Answer in its simplest form. | 2 |
|-----|--|---|

- | | | |
|-----|---|---|
| (g) | Find the values of x for which $ x + 2 \geq 5$. | 2 |
|-----|---|---|

Question 2 (12 marks)	Use a SEPARATE writing booklet	Marks
(a) Differentiate the following functions:		
(i) $y = x\sqrt{x}$		1
(ii) $y = x^3 \log_e x$		2
(iii) $y = \cos^2 x$		2
(iv) $y = \frac{x^3}{x^2 - 1}$		2
(b) Find $\int \sec^2 8x \, dx$		1
(c) Evaluate $\int_0^2 e^{3x} + 4 \, dx$		2
(d) Find the equation of the tangent to the curve $y = (1 + 2x)^3$ at the point (0,1).		2

Question 3 (12 marks) Use a SEPARATE writing booklet

The diagram shows the points $A(-1, 0)$ and $B(1, 6)$. $\angle BAC = \angle BCA$

Copy the diagram onto your examination paper.

- | | | |
|--------|---|---|
| (i) | Find the length of AB in its simplest surd form | 1 |
| (ii) | Find the midpoint of AB. | 1 |
| (iii) | Find the gradient of AB. | 1 |
| (iv) | Find the size of $\angle BAC$, correct to the nearest minute. | 1 |
| (v) | Find the equation of the perpendicular bisector of AB. | 2 |
| (vi) | Explain why the gradient of BC is -3 . | 1 |
| (vii) | Show that the equation of BC is $y = -3x + 9$. | 1 |
| (viii) | Find the coordinates of C. | 1 |
| (ix) | Find the perpendicular distance from A to BC. | 2 |
| (x) | Write down the co-ordinates of D such that BCAD is a parallelogram. | 1 |

Question 4 (12 marks) Use a SEPARATE writing booklet**Marks**

(a) The gradient function of a curve is given by $\frac{dy}{dx} = 3x^2 - 6x - 9$. The curve passes through the point $(1, -2)$.

- | | | |
|-------|---|----------|
| (i) | Find the equation of the curve. | 2 |
| (ii) | Find the co-ordinates of the stationary points and determine their nature. | 4 |
| (iii) | Find the co-ordinates of the point of inflexion. | 1 |
| (iv) | Sketch the curve showing the stationary points, point of inflexion and the y-intercept. | 2 |

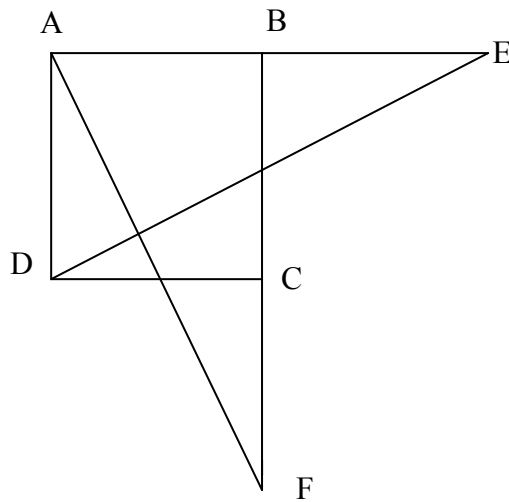
(b) The equation of a parabola is given by $(x - 2)^2 = 2y + 6$

- | | | |
|-------|--------------------------------------|----------|
| (i) | Find the co-ordinates of the vertex. | 1 |
| (ii) | Find the co-ordinates of the focus. | 1 |
| (iii) | Find the equation of the directrix. | 1 |

Question 5 (12 marks) Use a SEPARATE writing booklet		Marks
(a)	(i) Sketch the graph of $y = \frac{1}{x}$ and $x^2 = 8y$ on the same axes.	2
	(ii) Show the co-ordinates of the point of intersection of the two graphs are $(2, \frac{1}{2})$.	1
	(iii) Find the area of the region bounded by the x -axis and the curves from $x = 0$ to $x = 2e$.	3
(b)	(i) Draw the graphs of $y = 2\sin x$ and $y = \tan x$ for $0 \leq x \leq 2\pi$	2
	(ii) Use the graph to find to determine the number of solutions to $2\sin x = \tan x$.	1
	(iii) Find all the values of x where $0 \leq x \leq 2\pi$ are solutions to the equation $2\sin x = \tan x$.	3

Question 6 (12 marks) Use a SEPARATE writing booklet**Marks**

- (a) In the diagram ABCD is a square. AB is produced to E so that $AB = BE$ and BC is produced to F so that $BC = CF$



- (i) Copy the diagram on to your examination booklet.
 - (ii) Prove that $\triangle AED \equiv \triangle BFA$ **3**
 - (iii) Hence prove that $\angle AED = \angle BFA$ **1**
- (b) A five card hand is dealt from a regulation pack of cards. Find the probability of
- (i) a flush (all five cards have the same suit). **2**
 - (ii) four aces and any other cards. **2**
- (c) The mass, m , in grams of a radio active substance at the end of t years is given by $m = 10e^{-0.009t}$. Find
- (i) the mass, to the nearest gram, after 10 years. **1**
 - (ii) the rate at which the mass is decreasing after 10 years **1**
 - (iii) the number of years, to the nearest year, for the substance to reach its half-life. **2**

Question 7 (12 marks) Use a SEPARATE writing booklet

- (a) The lengths of the rung of a ladder increase uniformly from 40 cm in the top rung to 75 cm in the bottom rung. If 13.8 m of wood are used to make the rungs, many rungs are there? 2

- (b) An author writes a manuscript, so that on the first day he writes 54 pages, on the second day 36 pages and on each succeeding day he writes $\frac{2}{3}$ of the number of pages of the preceding day.

- (i) How many pages does he write on the 5th day? 1

- (ii) What is the maximum number of pages he will write? 1

- (c) Solve $\log_3 x + \log_3 (x+8) = 2$ 3

- (d) Given the quadratic equation $x^2 - (2+k)x + 3k = 0$, find the value of k if the product of the roots is four times the sum. 2

- (e) Consider the function given by $f(x) = \frac{1}{4+x^2}$ for $0 \leq x \leq 2$ and that you will be using Simpson's rule with 5 function values 3

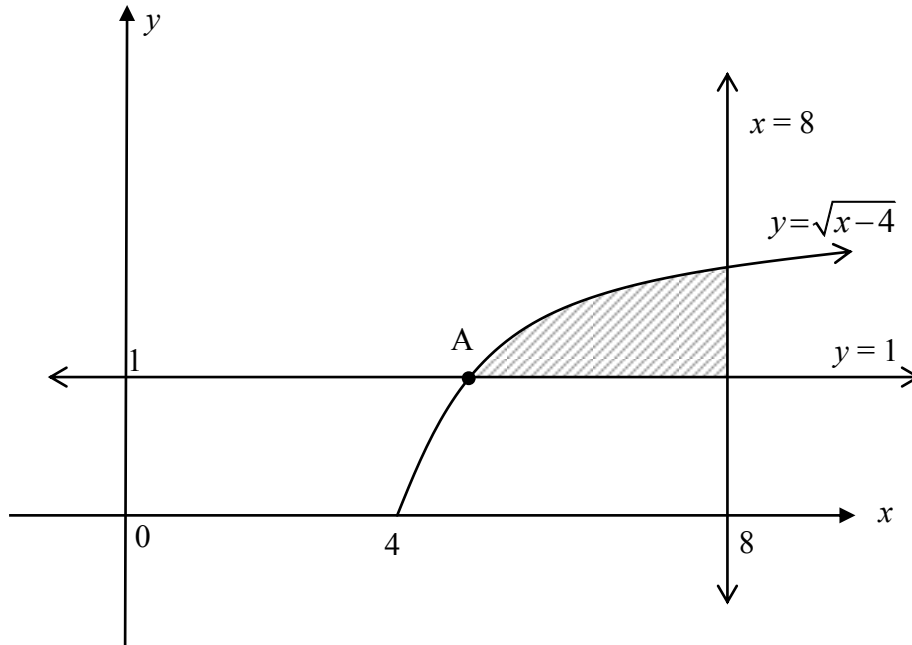
- (i) Copy and complete the table of values on your examination paper,

x					
$f(x)$					

- (ii) Find an approximation of $\int_0^2 f(x) dx$, correct to 3 decimal places.

Question 8 (12 marks) Use a SEPARATE writing booklet**Marks**

(a)



The region between $y = \sqrt{x-4}$, $y = 1$ and $x = 8$ is rotated about the x -axis.

- | | | |
|--|--|----------|
| (i) | Show that the co-ordinates of A are (5,1). | 1 |
| (ii) | Find the volume of the solid of revolution. | 3 |
|
(b) The position x cm of a particle P moving along the x -axis after t seconds is given by $x = 5 - 9t + 6t^2 - t^3$ | | |
| (i) | Find expressions for the particle's velocity and acceleration. | 2 |
| (ii) | In which direction does the particle initially move? | 1 |
| (iii) | When does the particle change direction? | 2 |
| (iv) | Where is the particle at these times? | 1 |
| (vi) | Find the total distance travelled between $t = 0$ and $t = 3$ | 2 |

Question 9 (12 marks) Use a SEPARATE writing booklet**Marks**

- (a) Brian borrows \$50000 at 18% p.a. compounded monthly and repays the loan in equal monthly instalments over 5 years.

- (i) Show that the amount owing after 2 months is

2

$$A_2 = 50000(1.015)^2 - M(1 + 1.015)$$

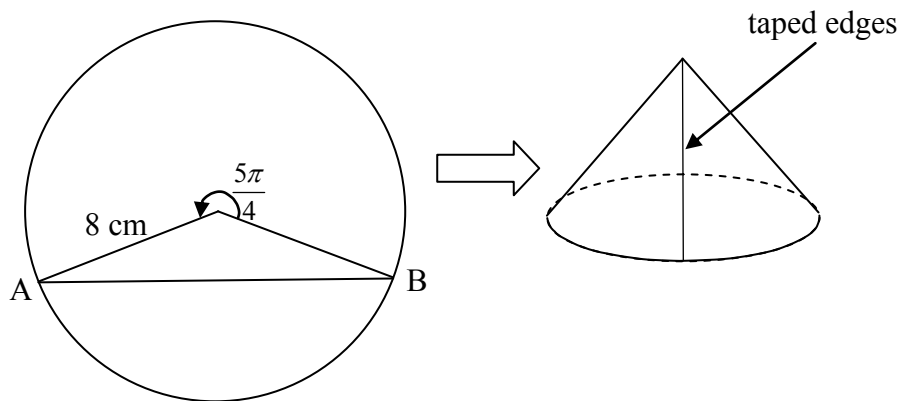
- (ii) Find the amount of each monthly repayment.

2

- (iii) How much interest does Brian pay?

2

(b)



- (i) Find the length of the minor arc AB. **1**
- (ii) Find the area of the minor sector AB. **1**
- (iii) Find the area of the minor segment AB. **2**
- (iv) If the major sector was taped to form a cone, what would be the radius of the base of the cone? **1**
- (v) What would be the surface area of the cone formed by joining A and B? **1**

Question 10 (12 marks) Use a SEPARATE writing booklet**Marks**

- (a) At 1:00 p.m. a ship A leaves port P and sails in the direction 030° T at 12 km/h .
Also at 1:00 p.m. ship B is 100 km due east of P and sailing at 8 km/h towards P.
Suppose t is the number of hours after 1p.m.

- (i) Show that the distance $D(t)$ between the 2 ships is given by **4**

$$D(t) = \sqrt{304t^2 - 2800t + 10000}$$

- (ii) Find the minimum value of $[D(t)]^2$ for all $t \geq 0$ **3**

- (iii) At what time, to the nearest minute, are the ships closest? **2**

- (b) Differentiate $y = \log_e \left[\frac{3x-1}{x+2} \right]$. **2**

- (c) Evaluate $y = \int \frac{e^{5x} + 4}{e^x} dx$. **1**

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

Question 1

$$\begin{aligned}
 \text{a) } 4\pi \sqrt{\frac{a}{9}} &= 4\pi \sqrt{\frac{4.2}{9.8}} \\
 &= 8.22662065 \\
 &= 8.2 \text{ to 2 sig fig}
 \end{aligned}$$

$$\text{b) } x+y = 2 \quad \text{--- ①}$$

$$2x-y = 7 \quad \text{--- ②}$$

$$\text{①} + \text{②} \quad 3x = 9$$

$$x = 3$$

$$\text{sub ①} \quad x+y = 2$$

$$3+y = 2$$

$$y = -1$$

$$\therefore x = 3, y = -1$$

$$\text{c) } 2x^2 - 3x - 14 = (2x-7)(x+2)$$

$$\text{d) } 180^\circ = \pi^\circ$$

$$1^\circ = \frac{\pi}{180}$$

$$135^\circ = 135 \times \frac{\pi}{180}$$

$$= \frac{3\pi}{4}$$

$$\text{e) } x^2 + 5^2 = (10-x)^2$$

$$x^2 + 25 = 100 - 20x + x^2$$

$$20x = 75$$

$$x = 3\frac{3}{4}$$

$$\text{f) } \frac{2}{\sqrt{5}-\sqrt{3}} = \frac{2}{\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}$$

$$= \frac{2(\sqrt{5}+\sqrt{3})}{5-3}$$

$$= \frac{2(\sqrt{5}+\sqrt{3})}{2}$$

$$= \sqrt{5} + \sqrt{3}$$

$$\text{g) } |x+2| \geq 5$$

$$x+2 \leq -5, \quad x+2 \geq 5$$

$$x \leq -7, \quad x \geq 3$$

Question 2

$$\text{a) i) } y = x\sqrt{x}$$

$$= x^{\frac{3}{2}}$$

$$\frac{dy}{dx} = \frac{3}{2} x^{\frac{1}{2}}$$

$$= \frac{3}{2} \sqrt{x}$$

$$\begin{aligned}
 \text{ii) } y &= x^3 \log_e x & \left| \begin{array}{l} u = x^3 \quad v = \log_e x \\ u' = 3x^2 \quad v' = \frac{1}{x} \end{array} \right. \\
 &= u v
 \end{aligned}$$

$$\frac{dy}{dx} = v u' + u v'$$

$$= 3x^2 \log_e x + x^3 \times \frac{1}{x}$$

$$= 3x^2 \log_e x + x^2$$

$$\text{iii) } y = \cos^2 x$$

$$= (\cos x)^2$$

$$\frac{dy}{dx} = -2 \cos x \sin x$$

$$\begin{aligned}
 \text{iv) } y &= \frac{x^3}{x^2-1} & \left| \begin{array}{l} u = x^3 \quad v = x^2-1 \\ u' = 3x^2 \quad v' = 2x \end{array} \right. \\
 &= \frac{u}{v}
 \end{aligned}$$

$$\frac{dy}{dx} = \frac{v u' - u v'}{v^2}$$

$$= \frac{3x^2(x^2-1) - x^3 \times 2x}{(x^2-1)^2}$$

$$= \frac{3x^4 - 3x^2 - 2x^4}{(x^2-1)^2}$$

$$= \frac{x^2(x^2-3)}{(x^2-1)^2}$$

$$b) \int \sec^2 8x \, dx = \frac{1}{8} \tan 8x + c$$

$$c) \int_0^2 e^{3x} + 4 \, dx = \left[\frac{1}{3} e^{3x} + 4x \right]_0^2$$

$$= \left[\frac{1}{3} e^6 + 8 \right] - \left[\frac{1}{3} e^0 \right]$$

$$= \frac{1}{3} e^6 + 7\frac{2}{3}$$

$$d) y = (1+2x)^3$$

$$\frac{dy}{dx} = 3(1+2x)^2 \times 2$$

$$= 6(1+2x)^2 \text{ at } x=0$$

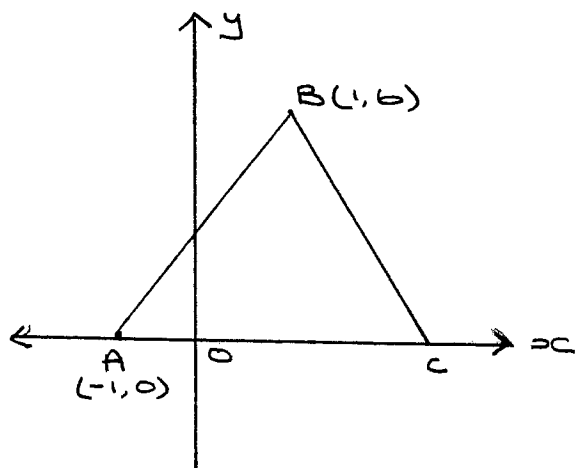
$$= 6$$

$$y - y_1 = m(x - x_1) \quad m=6, (0,1)$$

$$y - 1 = 6(x - 0)$$

$$y = 6x + 1$$

Question 3



$$i) AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(-1 - 1)^2 + (0 - 6)^2}$$

$$= \sqrt{4 + 36}$$

$$= \sqrt{40}$$

$$= 2\sqrt{10}$$

$$iii) \text{Midpoint } AB = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{-1 + 1}{2}, \frac{0 + 6}{2} \right)$$

$$= (0, 3)$$

$$ii) m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{6 - 0}{1 - -1}$$

$$= 3$$

$$\therefore \text{gradient } AB = 3$$

$$iv) m = \tan \theta$$

$$\hat{BAC} = \tan^{-1} 3$$

$$= 71^\circ 33' 54.18''$$

$$\hat{BAC} = 71^\circ 34' \text{ (nearest min)}$$

$$v) \text{Midpoint } AB = (0, 3)$$

$$\text{Gradient } AB = 3$$

$$\text{Gradient of required line} = -\frac{1}{3}$$

as $m_1 m_2 = -1$ for perpendicular lines

$$\therefore y - y_1 = m(x - x_1)$$

$$y - 3 = -\frac{1}{3}(x - 0)$$

$$3y - 9 = -x$$

$$x + 3y - 9 = 0$$

$$vi) \hat{BAC} = \hat{BCA}$$

Tangent ratio is negative for angles in the second quadrant

$$vii) (1, 6) \quad m = -3$$

$$y - y_1 = m(x - x_1)$$

$$y - 6 = -3(x - 1)$$

$$y - 6 = -3x + 3$$

$$y = -3x + 9$$

viii) $y = -3x + 9$ at C, $y = 0$

$$0 = -3x + 9$$

$$3x = 9$$

$$x = 3$$

$$\therefore C \text{ is } (3, 0)$$

ix) $3x + y - 9 = 0$ A(-1, 0)

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|3(-1) + 0 - 9|}{\sqrt{3^2 + 1^2}}$$

$$= \frac{|-12|}{\sqrt{10}}$$

$$= \frac{12}{\sqrt{10}} \text{ or } \frac{6\sqrt{10}}{5}$$

x) D is (-3, 6)

Question 4

a) i) $\frac{dy}{dx} = 3x^2 - 6x - 9$

$$y = \int (3x^2 - 6x - 9) dx$$

$$y = x^3 - 3x^2 - 9x + c \text{ at } (1, -2)$$

$$-2 = (1)^3 - 3(1)^2 - 9(1) + c$$

$$-2 = 1 - 3 - 9 + c$$

$$c = 9$$

$$\therefore y = x^3 - 3x^2 - 9x + 9$$

ii) For stationary points

$$\frac{dy}{dx} = 0$$

$$3x^2 - 6x - 9 = 0$$

$$3(x^2 - 2x - 3) = 0$$

$$3(x - 3)(x + 1) = 0$$

$$x = 3, -1$$

when $x = 3$

$$y = x^3 - 3x^2 - 9x + 9$$

$$= (3)^3 - 3(3)^2 - 9(3) + 9$$

$$= -18$$

$$\frac{d^2y}{dx^2} = 6x - 6 \text{ at } x = 3$$

$$= 6(3) - 6$$

$$= 12$$

As $\frac{d^2y}{dx^2} > 0$ the curve is concave up $\cup$

$\therefore (3, -18)$ is a local minimum

when $x = -1$

$$y = x^3 - 3x^2 - 9x + 9$$

$$= (-1)^3 - 3(-1)^2 - 9(-1) + 9$$

$$= 14$$

$$\frac{d^2y}{dx^2} = 6x - 6 \text{ at } x = -1$$

$$= 6(-1) - 6$$

$$= -12$$

As $\frac{d^2y}{dx^2} < 0$ the curve is concave down $\cap$

$\therefore (-1, 14)$ is a local maximum

iii) For points of inflexion

$$\frac{d^2y}{dx^2} = 0 \text{ and there is}$$

a change in concavity

$$\frac{d^2y}{dx^2} = 6x - 6 = 0$$

$$x = 1$$

when $x = 1$, $y = x^3 - 3x^2 - 9x + 9$

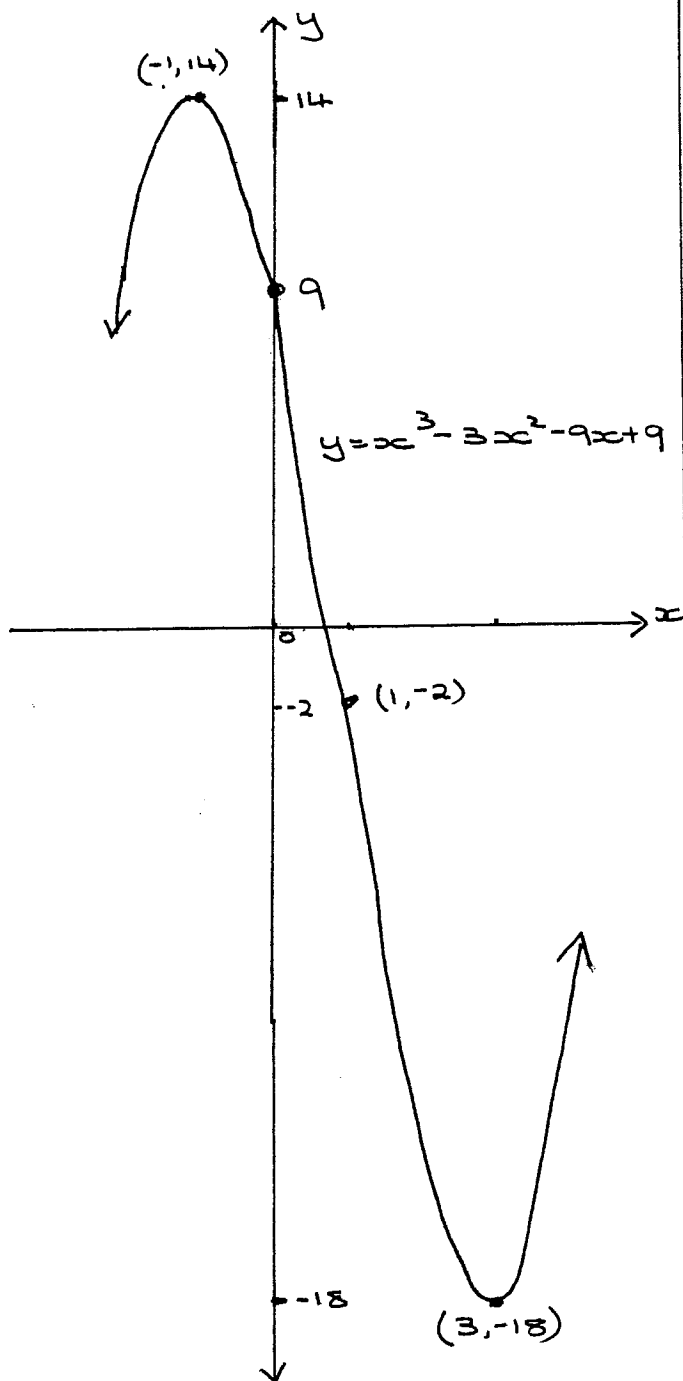
$$= (1)^3 - 3(1)^2 - 9(1) + 9$$

$$= -2$$

x	1 ⁻	1	1 ⁺
$\frac{d^2y}{dx^2}$	-	0	+

$\therefore$ change in concavity

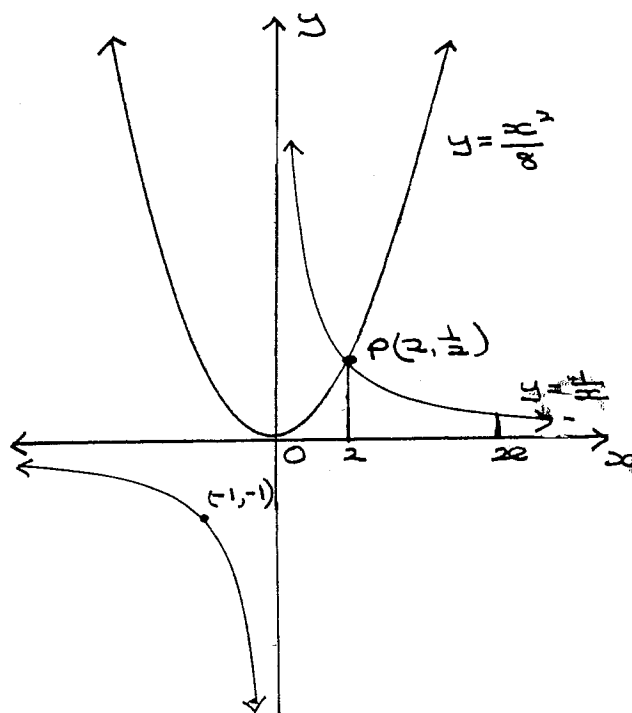
$\therefore (1, -2)$ is a point of inflexion



b) $(x-2)^2 = 2(y+3)$
 $4a = 2$
 $a = \frac{1}{2}$

- (i) vertex $(2, -3)$
- (ii) focus $(2, -2\frac{1}{2})$
- (iii) directrix $y = -3\frac{1}{2}$

Question 5



ii) $y = \frac{1}{x}$ — ①
 $y = \frac{x^3}{8}$ — ②

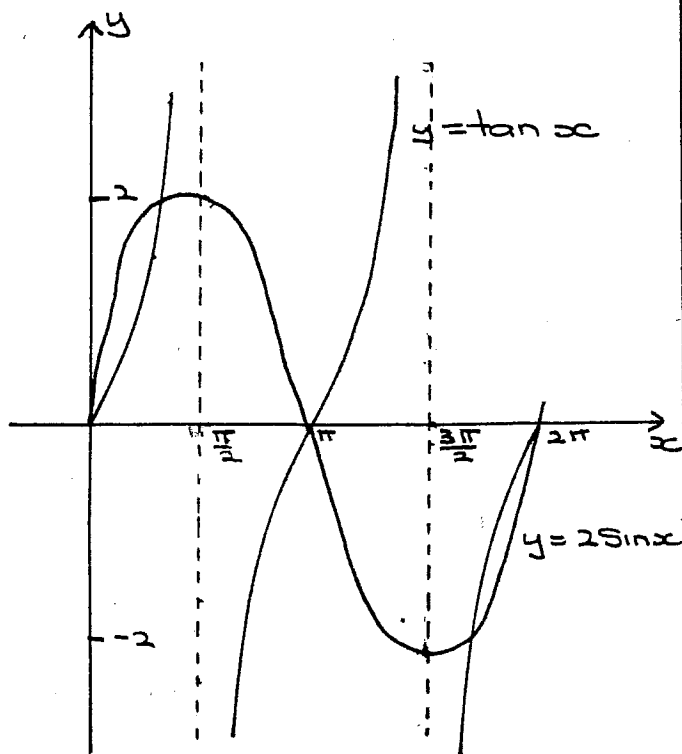
sub ① in ②
 $\frac{1}{x} = \frac{x^3}{8}$
 $x^3 = 8$
 $x = 2$

sub ① $y = \frac{1}{x}$
 $= \frac{1}{2}$

$\therefore (2, \frac{1}{2})$ is the point of intersection

iii) $A = \int_0^2 \frac{x^2}{8} dx + \int_2^{2e} \frac{1}{x} dx$
 $= \left[\frac{x^3}{24} \right]_0^2 + \left[\ln x \right]_2^{2e}$
 $= \left[\frac{2^3}{24} - 0 \right] + \left[\ln 2e - \ln 2 \right]$
 $= \frac{8}{24} + \ln 2 + \ln e - \ln 2$
 $= \frac{1}{3} + 1$

Area $= 1\frac{1}{3} \text{ units}^2$



(ii) $2 \sin x = \tan x$
has 5 solutions

$$\begin{aligned} \text{(iii)} \quad 2 \sin x &= \tan x \\ 2 \sin x - \tan x &= 0 \\ 2 \sin x - \frac{\sin x}{\cos x} &= 0 \end{aligned}$$

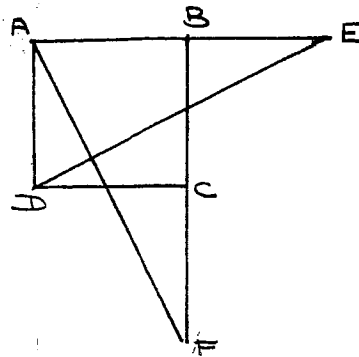
$$\begin{aligned} 2 \sin x \cos x - \sin x &= 0 \\ \sin x (2 \cos x - 1) &= 0 \end{aligned}$$

$$\begin{aligned} \sin x = 0 \quad \text{or} \quad \cos x = \frac{1}{2} \\ x = 0, \pi, 2\pi \quad x = \frac{\pi}{3}, \pi - \frac{\pi}{3} \end{aligned}$$

$$x = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

Question 6



In $\triangle AED$ and $\triangle BFA$

$AD = AB$ $ABCD$ is a square

$AE = BF$ $ABCD$ is a square

$AB = BE, BC = CF$

$\hat{DAE} = \hat{ABF}$ $ABCD$ is a square

$\therefore \triangle AED \equiv \triangle BFA$ SAS

$\therefore \hat{AED} = \hat{BFA}$

corresponding angles in congruent triangles

$$\begin{aligned} \text{b) } P(\text{flush}) &= 1 \times \frac{12}{51} \times \frac{11}{50} \times \frac{10}{49} \times \frac{9}{48} \\ &= \frac{11880}{5997600} \\ &= \frac{33}{16660} \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(4 \text{ aces}) &= \frac{4}{52} \times \frac{3}{51} \times \frac{2}{50} \times \frac{1}{49} \times 1 \\ &= \frac{24}{6497400} \\ &= \frac{1}{270725} \end{aligned}$$

$$\begin{aligned}
 \text{c) (i), } m &= 10e^{-0.009t} \quad t=10 \\
 &= 10e^{-0.009 \times 10} \\
 &= 10e^{-0.09} \\
 &\approx 9.139311853
 \end{aligned}$$

Mass = 9 grams to nearest gram

$$\begin{aligned}
 \text{(ii)} \quad \frac{dm}{dt} &= -0.009 \times 10e^{-0.009t} \\
 &= -0.09e^{-0.09} \\
 &\approx -0.082253806
 \end{aligned}$$

$\therefore$ Decreasing at the rate of 0.08g per year (to 2dp)

(iii) when $t=0$ $m_0=10$, $m=5$

$$\begin{aligned}
 m &= 10e^{-0.009t} \\
 5 &= 10e^{-0.009t} \\
 \frac{1}{2} &= e^{-0.009t} \\
 \ln\left(\frac{1}{2}\right) &= \ln(e^{-0.009t}) \\
 -0.009t &= \ln\left(\frac{1}{2}\right) \\
 t &= \frac{\ln\left(\frac{1}{2}\right)}{-0.009} \\
 &\approx 77.0163534
 \end{aligned}$$

The time to reach its half life is 77 years to nearest year

Question 7

a) Let there be n rungs
 $a=40$, $L=75$ $S_n=1380$

$$\begin{aligned}
 S_n &= \frac{n}{2}(a+L) \\
 1380 &= \frac{n}{2}(40+75) \\
 \frac{n}{2} &= \frac{1380}{115} \\
 &= 12 \\
 n &= 24
 \end{aligned}$$

$\therefore$ There are 24 rungs

$$\begin{aligned}
 \text{b) (i)} \quad a &= 54, \quad r = \frac{2}{3}, \quad n=5 \\
 T_n &= ar^{n-1} \\
 T_5 &= 54\left(\frac{2}{3}\right)^4 \\
 &= 10\frac{2}{3}
 \end{aligned}$$

$\therefore$ He writes $10\frac{2}{3}$ pages

$$\begin{aligned}
 \text{(ii)} \quad S_\infty &= \frac{a}{1-r} \quad |r| < 1 \\
 &= \frac{54}{1-\frac{2}{3}} \\
 &= 162
 \end{aligned}$$

$\therefore$ He will write 162 pages

$$\text{c) } \log_3 x + \log_3 (x+8) = 2$$

$$\begin{aligned}
 \log_3 x(x+8) &= 2 \\
 x(x+8) &= 3^2 \\
 x^2 + 8x - 9 &= 0 \\
 (x+9)(x-1) &= 0 \\
 x &= -9, 1
 \end{aligned}$$

But $x > 0 \therefore x=1$ is the only solution

$$\text{d) } x^2 - (2+k)x + 3k = 0$$

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha + \beta = 2+k \quad \text{--- (1)}$$

$$\alpha\beta = \frac{c}{a}$$

$$\alpha\beta = 3k \quad \text{--- (2)}$$

$$\alpha\beta = 4(\alpha + \beta)$$

$$3k = 4(2+k)$$

$$3k = 8 + 4k$$

$$k = -8$$

e) $f(x) = \frac{1}{4+x^2} \quad 0 \leq x \leq 2$

x	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2
$f(x)$	$\frac{1}{4}$	$\frac{4}{17}$	$\frac{1}{5}$	$\frac{4}{25}$	$\frac{1}{8}$

$$\int_a^b f(x) dx = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

$$\begin{aligned} \int_0^2 \frac{1}{4+x^2} dx &= \frac{2-0}{6} \left[\frac{1}{4} + \frac{1}{8} + 4\left(\frac{4}{17} + \frac{4}{25}\right) + \frac{1}{8} \right] \\ &= \frac{1}{6} [2.356176471 \dots] \\ &= 0.392696078 \\ &= 0.393 \text{ to 3dp} \end{aligned}$$

Question 8

a) i. $y = \sqrt{x-4}$ at $y=1$
 $1 = \sqrt{x-4}$
 $1 = x-4$
 $x = 5$
 $\therefore A$ is $(5, 1)$

ii) $V = \pi \int_a^b y^2 dx$
 $= \pi \int_5^8 (\sqrt{x-4})^2 dx = \pi \int_5^8 (x-4) dx$
 $= \pi \left[\frac{x^2}{2} - 4x \right]_5^8 = \pi \left[\frac{8^2}{2} - 4(8) - \left(\frac{5^2}{2} - 4(5) \right) \right]$
 $= 4\frac{1}{2}\pi$

$\therefore$ Volume is $4\frac{1}{2}\pi$ units³

b) $x = 5 - 9t + 6t^2 - t^3$

i) $\dot{x} = -9 + 12t - 3t^2$
 $\ddot{x} = 12 - 6t$

ii) Initial $\dot{x} = -9 \text{ cm s}^{-1}$

That is moving with a velocity of 9 cm s^{-1} is the negative direction $\leftarrow$

iii) Particle changes direction when $\dot{x} = 0$

$$-9 + 12t - 3t^2 = 0$$

$$t^2 - 4t + 3 = 0$$

$$(t-3)(t-1) = 0$$

$$t = 1, 3$$

Particle changes direction after 1 second and 3 seconds

(iv) When $t = 1$

$$\begin{aligned} x &= 5 - 9t + 6t^2 - t^3 \\ &= 5 - 9(1) + 6(1)^2 - (1)^3 \\ &= 5 - 9 + 6 - 1 \\ &= 1 \end{aligned}$$

when $t = 3$

$$\begin{aligned} x &= 5 - 9t + 6t^2 - t^3 \\ &= 5 - 9(3) + 6(3)^2 - 3^3 \\ &= 5 \end{aligned}$$

when $t = 1$ the particle is 1cm to the right of 0 and when $t = 3$ the particle is 5cm to the right of 0

t	0	1	2	3
x	5	1	3	5

Total distance = $4 + 2 + 2 = 8 \text{ cm}$

Question 9

a) let A_n be the amount owing at the end of the n^{th} month

let M be the monthly repayment

$$\begin{aligned} n &= 5 \times 12 \\ &= 60 \end{aligned} \quad \begin{aligned} r &= \frac{18}{12} \% \\ &= 1.5 \% \\ &= 0.015 \end{aligned}$$

$$\begin{aligned} A_1 &= 50000 + 1.5 \% \text{ of } 50000 - M \\ &= 50000 + 0.015 \times 50000 - M \\ &= 50000(1.015) - M \end{aligned}$$

$$\begin{aligned} A_2 &= A_1 + 1.5 \% \text{ of } A_1 - M \\ &= A_1 + 0.015 \times A_1 - M \\ &= A_1(1.015) - M \\ &= [50000(1.015) - M](1.015) - M \\ &= 50000(1.015)^2 - M(1 + 1.015) \end{aligned}$$

continuing the pattern

$$A_{60} = 50000(1.015)^{60} - M(1 + 1.015 + \dots + 1.015^{59})$$

But after 60 months the loan is repaid $\therefore A_{60} = 0$

$$50000(1.015)^{60} - M(1 + 1.015 + \dots + 1.015^{59}) = 0$$

$$\begin{aligned} M &= \frac{50000(1.015)^{60}}{1 + 1.015 + \dots + 1.015^{59}} \\ &= \frac{50000(1.015)^{60}}{\frac{1(1.015^{60} - 1)}{1.015 - 1}} \end{aligned}$$

$$= \frac{50000(1.015)^{60}(0.015)}{(1.015^{60} - 1)}$$

$$= 1269.671371$$

$$= 1269.67 \text{ to 2 dp}$$

$\therefore$ Each monthly instalment is \$1269.67

$$\begin{aligned} \text{(iii)} \quad I &= 1269.67 \times 60 - 50000 \\ &= 26180.2 \end{aligned}$$

$\therefore$ Interest paid is \$26180.20

$$\begin{aligned} \text{b) (i)} \quad L &= r\theta \quad r = 8 \quad \theta = \frac{3\pi}{4} \\ &= 8 \times \frac{3\pi}{4} \end{aligned}$$

$$= 6\pi$$

length of the arc is 6π cm

$$\begin{aligned} \text{(ii)} \quad A &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} 8^2 \times \frac{3\pi}{4} \\ &= 24\pi \end{aligned}$$

Area is $24\pi \text{ cm}^2$

$$\begin{aligned} \text{(iii)} \quad A &= \frac{1}{2} r^2 (\theta - \sin \theta) \\ &= \frac{1}{2} \times 8^2 \left(\frac{3\pi}{4} - \sin \frac{3\pi}{4} \right) \end{aligned}$$

$$= 32 \left(\frac{3\pi}{4} - \frac{1}{\sqrt{2}} \right)$$

$$= 24\pi - 16\sqrt{2}$$

$$= 8(3\pi - 2\sqrt{2})$$

$\therefore$ Area is $8(3\pi - 2\sqrt{2}) \text{ cm}^2$
(or 52.77680669 cm^2)

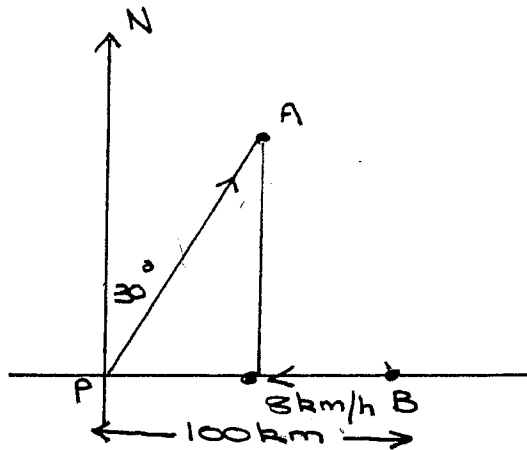
$$\begin{aligned} \text{iv) Major Arc} &= r\theta \\ &= 8 \times \frac{5\pi}{4} \\ &= 10\pi \\ 2\pi r &= 10\pi \\ r &= 5 \end{aligned}$$

$\therefore$ Radius is 5 cm

$$\begin{aligned} \text{v) } S.A &= \pi r L \\ &= \pi \times 5 \times 8 \\ &= 40\pi \end{aligned}$$

Surface area is $40\pi \text{ cm}^2$

Question 10



$$\begin{aligned} \text{At time } t, \quad PA &= 12t \\ PB &= 100 - 8t \end{aligned}$$

Using cosine rule

$$\begin{aligned} p^2 &= a^2 + b^2 - 2ab \cos P \\ AB^2 &= (100 - 8t)^2 + (12t)^2 \\ &\quad - 2(100 - 8t)(12t) \cos 60 \end{aligned}$$

$$\begin{aligned} &= 10000 - 1600t + 64t^2 + 144t^2 \\ &\quad - 1200t + 96t^2 \end{aligned}$$

$$= 10000 - 2800t + 304t^2$$

$$\therefore D(t) = \sqrt{304t^2 - 2800t + 10000}$$

$$\text{(ii) } [D(t)]^2 = 304t^2 - 2800t + 10000$$

$$\frac{d[D(t)]^2}{dt} = 608t - 2800$$

$$\begin{aligned} 608t - 2800 &= 0 \\ t &= 4 \frac{23}{38} \end{aligned}$$

$$\begin{aligned} [D(t)]^2 &= 10000 - 2800t + 304t^2 \\ &= 10000 - 2800\left(4 \frac{23}{38}\right) + 304\left(4 \frac{23}{38}\right)^2 \\ &\approx 3552.631579 \\ &= 3553 \text{ to nearest km} \end{aligned}$$

t	$4 \frac{23}{38}^-$	$4 \frac{23}{38}$	$4 \frac{23}{38}^+$
$\frac{[D(t)]^2}{dt}$	-	0	+

$\therefore$ Minimum value of $[D(t)]^2$ is 3553

$$\begin{aligned} \text{(iii) } 1 \text{ pm} &+ 4 \frac{23}{38} \text{ hr} \\ &\approx 1 \text{ pm} + 4 \text{ hr } 36 \text{ min} \\ &= 5:36 \text{ pm} \end{aligned}$$

$$\begin{aligned} \text{b) } y &= \log_e \left(\frac{3x-1}{x+2} \right) \\ &= \log_e(3x-1) - \log_e(x+2) \\ \frac{dy}{dx} &= \frac{3}{3x-1} - \frac{1}{x+2} \\ &= \frac{3(x+2) - (3x-1)}{(3x-1)(x+2)} \\ &= \frac{7}{(3x-1)(x+2)} \end{aligned}$$

$$\begin{aligned} \text{c) } y &= \int \left(\frac{e^{5x} + 4}{e^x} \right) dx \\ &= \int (e^{4x} + 4e^{-x}) dx \\ &= \frac{1}{4} e^{4x} - 4e^{-x} + C \end{aligned}$$